

Indian Institute of Technology Dharwad
CS201L: Artificial Intelligence Laboratory
Lab 2: Data Visualization and Descriptive analytics
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Problem Statement 1

A residential housing dataset is provided in a CSV file (`residential_housing.csv`). The dataset contains various attributes that describe the physical, locational, and quality-related characteristics of houses. These features are:

Table 1: Description of Attributes in the Residential Housing Dataset

Feature	Description
Id	Unique identifier assigned to each property record.
MSSubClass	Type of dwelling involved in the transaction.
MSZoning	General zoning classification of the property.
LotFrontage	Linear feet of street connected to the property.
LotArea	Total lot size in square feet.
Street	Type of road access to the property.
Alley	Type of alley access to the property.
LotShape	General shape of the property lot.
LandContour	Flatness of the property.
Utilities	Type of utilities available at the property.
LotConfig	Lot configuration.
LandSlope	Slope of the property.
Neighborhood	Physical location within the city.
Condition1	Proximity to main road or railroad.
Condition2	Secondary proximity to main road or railroad.
BldgType	Type of dwelling.
HouseStyle	Style of dwelling.
OverallQual	Overall material and finish quality.
OverallCond	Overall condition rating.
YearBuilt	Original construction year.
YearRemodAdd	Year of remodeling or addition.
RoofStyle	Type of roof.
RoofMatl	Roof material.
Exterior1st	Primary exterior covering.

Feature	Description
Exterior2nd	Secondary exterior covering.
MasVnrType	Masonry veneer type.
MasVnrArea	Masonry veneer area in square feet.
ExterQual	Exterior material quality.
ExterCond	Exterior material condition.
Foundation	Type of foundation.
BsmtQual	Basement height quality.
BsmtCond	Basement condition.
BsmtExposure	Basement exposure level.
BsmtFinType1	Primary basement finished area type.
BsmtFinSF1	Size of primary finished basement area.
BsmtFinType2	Secondary basement finished area type.
BsmtFinSF2	Size of secondary finished basement area.
BsmtUnfSF	Unfinished basement area.
TotalBsmtSF	Total basement area.
Heating	Type of heating.
HeatingQC	Heating quality and condition.
CentralAir	Central air conditioning availability.
Electrical	Electrical system type.
1stFlrSF	Area of first floor in square feet.
2ndFlrSF	Area of second floor in square feet.
LowQualFinSF	Low quality finished living area.
GrLivArea	Above ground living area.
BsmtFullBath	Number of basement full bathrooms.
BsmtHalfBath	Number of basement half bathrooms.
FullBath	Number of full bathrooms above ground.
HalfBath	Number of half bathrooms above ground.
BedroomAbvGr	Number of bedrooms above ground.
KitchenAbvGr	Number of kitchens above ground.
KitchenQual	Kitchen quality rating.
TotRmsAbvGrd	Total number of rooms above ground.
Functional	Home functionality rating.
Fireplaces	Number of fireplaces.
FireplaceQu	Fireplace quality.
GarageType	Garage location type.
GarageYrBlt	Year garage was built.
GarageFinish	Garage interior finish.
GarageCars	Number of vehicles the garage can hold.
GarageArea	Size of garage in square feet.

Feature	Description
GarageQual	Garage quality.
GarageCond	Garage condition.
PavedDrive	Type of paved driveway.
WoodDeckSF	Wood deck area in square feet.
OpenPorchSF	Open porch area.
EnclosedPorch	Enclosed porch area.
3SsnPorch	Three-season porch area.
ScreenPorch	Screen porch area.
PoolArea	Pool area in square feet.
PoolQC	Pool quality rating.
Fence	Fence quality.
MiscFeature	Miscellaneous feature not covered in other categories.
MiscVal	Value of miscellaneous feature.
MoSold	Month of sale.
YrSold	Year of sale.
SaleType	Type of sale.
SaleCondition	Condition of sale.
SalePrice	Recorded sale price of the property.

Write a Python program to perform the following tasks:

1. Generate descriptive statistics using the `describe()` function
2. Compute the first quartile (Q_1), median (Q_2), third quartile (Q_3), and the interquartile range (IQR) for the `LotArea` attribute using the `quantile()` function.
3. Using the Interquartile Range (IQR) rule, determine the lower and upper bounds for the `LotArea` attribute. Let Q_1 and Q_3 denote the first and third quartiles of `LotArea`, respectively. The bounds are computed as:

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper Bound} = Q_3 + 1.5 \times \text{IQR},$$

where $\text{IQR} = Q_3 - Q_1$. Compute these bounds using the `quantile()` function.

4. Among the attributes `LotArea`, `GrLivArea`, and `SalePrice`, determine the attribute having the highest interquartile range (IQR). Compute the IQR for each attribute using the `quantile()` function and identify the attribute with the maximum IQR value.
5. Generate scatter plots to explore the relationships between `SalePrice`, `OverallQual`, `GrLivArea`, and `YearBuilt` using both Matplotlib and Seaborn libraries.
6. Generate histogram and KDE visualizations for the variables `SalePrice`, `LotArea`, and `OverallQual` using both Matplotlib and Seaborn frameworks. For each distribution, overlay vertical reference lines indicating the mean, median, and ± 2.7 standard deviation limits, and annotate these values directly on the plots to facilitate statistical interpretation.

7. Group the dataset by the attribute `OverallQual` and generate bar plots for `SalePrice`, `GrLivArea`, and `YearBuilt` using both Matplotlib and Seaborn libraries.
8. Create boxplots for all numerical attributes to identify outliers and compare distributions using both Matplotlib and Seaborn libraries.